NRR-Guarantee: Bounded Commit/Defer Guarantee Surfaces and Explicit Downstream Boundaries

Kei Saito*
Independent Researcher, Japan
kei.saito.research@gmail.com

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Abstract

This paper studies a bounded guarantee question for commit/defer behavior under explicit provider, family, horizon, and observability conditions. It reports one supported early ambiguity-turn result together with downstream-acceptance, side-topic, and later-horizon limits. In the tested provider pair, the early ambiguity-turn comparison moves from `0 / 24` to `24 / 24` on both providers for `resolution_attempt_present` and `appropriate`, and from `8 / 24` to `24 / 24` for `unilateral_commit_absent`. At the downstream-acceptance boundary, the matched `baseline vs Energy policy` comparison supports bounded `preacceptance_rework_burden_absent` improvement: both provider slices move from `0 / 24` to `24 / 24`, while Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`. The auxiliary repaired `NRR vs Ablation` comparison remains mixed (`Anthropic = inconclusive`, `Gemini = support`; Anthropic `Baseline` remains stronger than Anthropic `NRR` on `appropriate`). The repaired first-step side-topic result reaches `12 / 12` on both providers, but it is reported only as supporting evidence because it depends on a local `baseline vs revised Energy policy` comparison rather than

*ORCID: 0009-0006-4715-9176

on the paper's matched `baseline vs Energy policy` comparison. Taken together, the evidence supports one narrow guarantee claim with explicit downstream, side-topic, and later-horizon limits rather than a provider-universal or long-horizon guarantee.

1 Introduction

NRR is not an anti-LLM framework. NRR does not replace standard LLM use. NRR optimizes when to commit and when to defer, under explicit conditions.

Related work in mixed-initiative and task-oriented dialogue has long examined adaptive initiative control and goal-oriented interaction management [8, 10], while clarification-focused lines study how a system should ask follow-up questions when a user's request remains ambiguous or underspecified [9, 11, 12]. Grounding-oriented work further emphasizes that dialogue systems must recover collaboratively when interpretation is not yet fixed [13, 14]. More recent LLM-centered work has also revisited clarification under query ambiguity and abstention or withholding under uncertainty [15, 16, 17]. Work on verified AI and runtime assurance frames a complementary question: what scoped assurances can be justified under explicit assumptions, monitors, and failure boundaries rather than by unconstrained behavioral optimism [18, 19]. This paper uses `Guarantee` in that bounded verification/assurance sense: not as a theorem-level or provider-universal correctness claim, but as a conditioned claim whose provider scope, horizon scope, observability scope, and failure boundaries are made explicit and auditable.

Within the NRR series, `Core` and `Phi` fixed the upstream position-dependent identity and state-mapping results [1, 2]. `IME`, `Transfer`, and `Coupled` then constrained the interface, transfer, and dependency conditions [3, 4, 5]. Integrated `paper7` fixed the delayed-commitment comparison and boundary-honesty basis, and `Energy` fixed the calibration/control basis that the present manuscript inherits directly [6, 7].

This paper uses `Policy`, `Dynamics`, and `Policy-Verification` as supporting inputs for a bounded `Guarantee` claim. Its purpose is to state only those downstream conclusions that remain supportable when the dependency chain is kept at `paper7` → `Energy` → `Guarantee` and when provider, family, horizon, observability, and failure-boundary limits are made explicit.

The Core-to-Guarantee program goal itself is not being rewritten here. What the series must establish from `Core` through `Guarantee` remains the same: a framework grounded in position-dependent identity as a first prin-

ciple, closed as an engineering program of conditional utility through implementation, quality measurement, operational control, and verification. The current *non-evaluative principle* read reorganizes that same program; it does not replace the top-level goal. In this paper, the practical use of that read is to tighten closure discipline around a narrow notion of 'evaluation', namely full branchwise comparative evaluation during retention.

The scope is intentionally narrow. The paper states only the bounded claim that the current evidence supports, without promoting any provider-universal, family-universal, or long-horizon statement.

Series Consistency Statement: Unsupported interpretations are never injected at a fixed evidence state, hypothesis-set expansion is evidence-gated, and unobserved possibilities remain explicitly unresolved. In this paper, the evidence-source clause is restricted to the supporting analyses and observability sources used for the present `Guarantee` claim.

Series Extensibility Statement: NRR is a derivation-and-evaluation framework, not a closed recipe. The evaluated policy/operator set is treated as an evaluated set under tested conditions, not as an exhaustive set. The aim of this paper is not to claim that these runs exhaust the guarantee question, but to provide one bounded claim with explicit supporting evidence, explicit failure boundaries, and comparison rules that later work can test under new providers, new families, and longer horizons.

2 Contributions

Under the currently fixed provider, family, horizon, and observability bounds, this manuscript makes four bounded contributions:

1. it establishes bounded early ambiguity-turn support for non-unilateral priority resolution under matched `baseline vs Energy policy` conditions;

2. it carries downstream acceptance as an explicit boundary surface: the mainline `baseline vs Energy policy` comparison supports shared `preacceptance_rework_burden_absent`, while visible provider-sliced residuals prevent pair-wide downstream-quality closure;

3. it carries repaired first-step side-topic / quarantine evidence as a bounded supporting surface under matched `baseline vs revised`

`Energy policy` conditions, with the main incremental lift concentrated in `C_incident_response` and without upgrading that local repaired comparison into a second manuscript supported claim;

4. it keeps later-horizon degradation and unsupported extensions explicit as failure-boundary reporting and reusable rerun surfaces rather than inflating them into broader guarantee language.

3 Program Position And Scope

This paper sits at the end of the current downstream chain:

$$\texttt{paper7} \rightarrow \texttt{Energy} \rightarrow \texttt{Guarantee}$$

The dependency split is fixed for interpretation. Integrated `paper7` supplies the comparison-and-boundary-honesty basis. `Energy` supplies the calibration/control basis and the imported evidence used here. `Policy` supplies the minimal state/action specification. `Dynamics` supplies the hold/degrade/break vocabulary over later checkpoints. `Policy-Verification` supplies the prospective matched- condition validation evidence. `Guarantee` then states the bounded supported, supporting, and boundary results that remain after those inputs are combined under explicit conditions. Figure 1 summarizes that dependency structure.

This means the manuscript has a double obligation. It must absorb the already fixed supporting layers without reopening their ownership boundaries, and it must state only those guarantee-shaped claims that remain supportable after explicit restriction of provider, family, horizon, observability, and failure boundary.

The current organizing read is likewise narrow. The *non-evaluative principle* is used here only as a local way to organize what counts as satisfactory closure for this manuscript, not as a theorem-level claim that supersedes the Core-to-Guarantee objective itself. In that narrower sense, `Guarantee` is the downstream line where the bounded supported, supporting, and boundary results must be stated without inflating them into unrestricted runtime branchwise comparison.

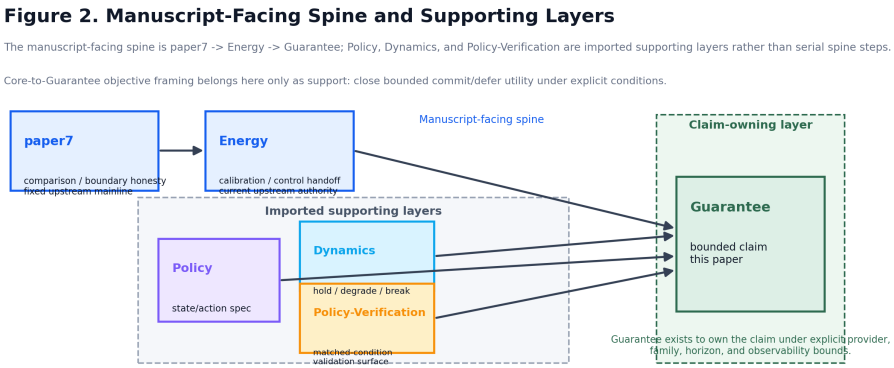

Figure 1: Dependency structure for the current `Guarantee` paper. Integrated `paper7` and `Energy` remain the direct upstream inputs, while `Policy`, `Dynamics`, and `Policy-Verification` are supporting inputs rather than serial upstream steps. `Guarantee` carries the bounded supported, supporting, and boundary results stated here.

4 Supporting Inputs

4.1 Policy Input

The current policy layer fixes the downstream state vocabulary (`active_objective`, `suspended_objective`, `accepted_direction`, `unresolved_relation`, `quarantine`, `local_relation_block`) together with the minimal commit/defer action set. The `Guarantee` line inherits this vocabulary as its operating language and does not redesign it locally.

That vocabulary is not ornamental. The early ambiguity-turn surface is the cleanest current instance of `defer_with_resolution`: no content work should begin before a direction is accepted, and no unilateral priority choice should be made on the user's behalf. The downstream acceptance surface is the current instance of `commit_to_accepted_direction` in which the honest carry-forward read is bounded support for `preacceptance_rework_burden_absent` under the mainline comparison contract, with Gemini residuals still visible beyond that shared gain rather than broad downstream closure. The repaired first-step side-topic surface is the clearest first-step instance of `accepted_direction` becoming `active_objective` while the prior line is held as `suspended_objective` under `quarantine`, but in the current paper it is carried only as a supporting surface because that read

depends on a local repaired comparison rather than on the manuscript's inherited mainline comparison contract. The later-horizon boundary then marks the point where `unresolved_relation` handling weakens and `local_relation_block` is no longer treated conservatively. For the current downstream line, preserving an unresolved relation should also be read in a narrower but more practical sense than mere recommendation avoidance. Once a branch has already been accepted, failure can still occur if the next assistant move silently compresses still-live user-side constraint or satisfaction structure into one asserted local reading. The manuscript does not expand that point into a general hidden-needs thesis. It uses it only to mark a bounded downstream failure mode: advancing as though the user's governing criterion were already fixed when the visible interaction state does not yet support that fixation.

4.2 Dynamics Input

The current dynamics layer contributes mechanism vocabulary only. Its role is to name where the inherited policy state holds, weakens, or breaks across later checkpoints. In the current branch, the key motifs are *Stable Resolution*, *Clean Quarantine Hold*, *Useful Continuation Hold*, *Relation-Preservation Hold*, *Premature Relation Collapse*, *Repeated-Starter Attractor*, *Generic Stall*, and *Quarantine Leak*.

Using that vocabulary makes the manuscript read as one bounded progression rather than as three disconnected pilot wins. The ambiguity-turn result is the clearest instance of *Stable Resolution*. The repaired first-step side-topic supporting surface is the clearest instance of *Clean Quarantine Hold*. The downstream acceptance result remains narrower: under the manuscript's inherited `baseline vs Energy policy` comparison, the shared supported behavior is bounded removal of `preacceptance_rework_burden_absent`, but Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`, and the auxiliary repaired `NRR vs Ablation` read remains mixed, with Anthropic `Baseline` still stronger than Anthropic `NRR` on `appropriate`. That surface should not yet be collapsed into uniform *Useful Continuation Hold*. Because the repaired side-topic read depends on a local repaired comparison, it remains supporting evidence rather than a second supported claim. The later-horizon boundary is best read as weakening *Relation-Preservation Hold* before *Premature Relation Collapse* appears, with more local residual forms such as *Generic Stall* or *Repeated-Starter Attractor* where the template/provider slice warrants them.

4.3 Policy-Verification Input

The current policy-verification layer fixes the protocol rules that any prospective `baseline vs Energy policy` comparison must satisfy before guarantee text is allowed. This includes matched-condition comparison, pre-frozen reporting surfaces, narrow family selection, explicit horizon bands, and main-text reporting of both gains and failure boundaries.

4.4 Experimental Conditions

The carried surfaces in this manuscript are reported as frozen matched-condition row totals rather than as claims of independent stochastic sampling. The early ambiguity-turn pilot uses temperature `0.0`, three templates, and eight reps per provider/arm, so each provider slice is reported over `24` rows at one early decision boundary. The downstream-acceptance pilot also uses temperature `0.0`, the same three templates, and four reps per acceptance variant across two acceptance variants, again yielding `24` rows per provider slice at the first accepted-direction boundary. The repaired first-step side-topic / quarantine pilot uses temperature `0.0`, the same three templates, `accept_side_first` only, and four reps per provider/arm, yielding `12` rows per provider slice at `turn5`. The later-horizon boundary then reports later checkpoints `turn7`, `turn9`, and `turn11` over `12` episodes per provider in a separate frozen `priority-resolution horizon-stability` family that reads the later continuation of the carried side-topic line. The numerators and denominators in the manuscript should therefore be read as fixed-condition row counts under frozen prompts, models, and templates, not as provider-universal frequency estimates.

5 Claim And Boundary Discipline

Any supported claim, supporting result, or explicit boundary result in this paper must disclose a bounded reporting specification with the following fields:

1. provider scope
2. family scope
3. horizon scope
4. observability scope

5. comparison scope
6. supported behavior
7. explicit failure boundary
8. unsupported extension

These fields define a shared disclosure schema, not one uniform evidentiary contract. Supported claims, supporting results, and matched-condition boundary results must fill all eight fields explicitly. The later-horizon boundary is instead carried as a diagnostic one-arm later-checkpoint degradation result: it still states provider, family, horizon, observability, comparison, supported behavior, explicit failure boundary, and unsupported extension, but its comparison scope is explicitly a diagnostic one-arm later-checkpoint read rather than a matched-comparison fill, and its supported-behavior field remains boundary-only rather than an additional supported claim. No scientific result in this paper is valid if the fields required by its evidentiary contract are omitted. This paper therefore inherits the current schema rule that the claim must be chosen before outcome inspection and must remain narrow enough to prevent provider-universal, family-universal, or long-horizon overreach. Table 1 summarizes the supported, supporting, and boundary results under that disclosure schema, and Figure 2 shows how those results relate across the current progression.

Unit	**Scope**	**Supported behavior**	**Boundary kept explicit**
Early ambiguity-turn	Tested provider pair, frozen priority-resolution family, early decision boundary	Non-unilateral priority resolution	No downstream acceptance, side-topic/quarantine, or later-horizon extension
Downstream acceptance boundary	Tested provider pair, priority-resolution family, first accepted-direction boundary	Bounded support for preacceptance burden removal under matched baseline vs Energy-policy conditions	Keep Gemini residuals and the auxiliary repaired read explicit; do not collapse into pair-wide or provider-universal downstream-guarantee language
Side-topic/quarantine supporting surface	Tested provider pair, repaired side-topic family, first accepted side-topic workstart checkpoint	Clean first-step side-topic workstart under explicit side-first acceptance on a local repaired comparison	Keep it as supporting evidence only; do not upgrade the local repaired comparison into a second manuscript supported claim or later-horizon stability read
Later-horizon boundary	Tested provider pair, priority-resolution horizon-stability family, later checkpoints `turn7`/`turn9`/`turn11`, diagnostic one-arm later-checkpoint read	Boundary-only read: relation preservation weakens before suspended-main quarantine breaks	Keep later-horizon as diagnostic boundary reporting only; no matched-comparison fill, no fourth supported claim, and no provider-universal or long-horizon extension

Table 1: Current supported, supporting, and boundary surfaces in `Guarantee`.

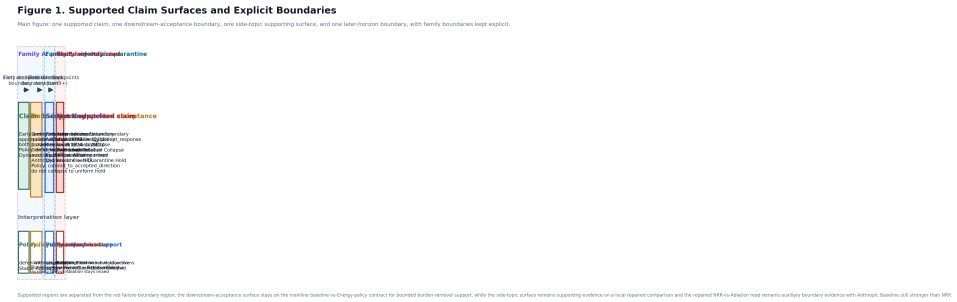

Figure 2: Supported claim progression, supporting evidence, and explicit boundary surfaces. Claim 1 and the downstream-acceptance boundary occupy different horizon points within the same `priority-resolution` family, with the downstream-acceptance surface marking the first accepted-direction boundary inside that family, while the side-topic/quarantine read belongs to a separate repaired family and is carried only as supporting evidence. The later-horizon region is a separate `priority-resolution horizon-stability` boundary read over the later continuation of that carried line rather than an additional supported claim.

5.1 First Supported Claim Unit

The first explicit supported claim unit is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the frozen `priority-resolution` ambiguity-turn family only
- horizon scope: the early decision boundary only
- observability scope: `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`
- comparison scope: matched baseline vs Energy-policy under the frozen early ambiguity-turn pilot conditions
- supported behavior: non-unilateral priority resolution
- explicit failure boundary: no downstream acceptance, side-topic/quarantine, or later-horizon closure is established by this fill

- unsupported extension: no provider-universal, long-horizon, or broad downstream-quality claim is authorized

In manuscript form, the bounded claim is:

> Under the tested provider pair, for the frozen `priority-resolution` ambiguity-turn family, at the early decision boundary, and on the directly scored fields `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`, the Energy-policy arm is supported relative to matched baseline only for non-unilateral priority resolution. This support excludes downstream acceptance, side-topic/quarantine, and later-horizon behavior, and it does not extend to provider-universal or broad downstream-quality claims.

5.2 Downstream-Acceptance Boundary Surface

The downstream-acceptance boundary surface is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the carried `priority-resolution` family only
- horizon scope: the first accepted-direction boundary inside that carried family only
- observability scope: `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`
- comparison scope: matched baseline vs Energy-policy under the frozen downstream-boundary pilot conditions, with reviewed `NRR vs Ablation` retained only as an auxiliary repaired boundary comparator
- supported behavior: bounded support for removal of preacceptance rework burden under the first accepted-direction downstream boundary
- explicit failure boundary: the surface does not close as pair-wide downstream quality, provider-universal continuation support, explicit quarantine closure, or later-horizon continuation stability; Gemini `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`, and the auxiliary repaired read remains mixed

- unsupported extension: no pair-wide downstream-acceptance guarantee, explicit quarantine closure, or later-horizon stability claim is authorized, and no manuscript claim that the auxiliary repaired `NRR vs Ablation` read replaces the paper-level comparison contract

  In manuscript form, the bounded boundary read is:

  > Under the tested provider pair, within the carried `priority-resolution` family, at the first accepted-direction boundary, and on the directly scored fields `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`, the matched `baseline vs Energy policy` surface supports only bounded removal of preacceptance rework burden rather than pair-wide downstream-quality closure: both provider slices reach `24 / 24` on `preacceptance_rework_burden_absent`, but Gemini `accepted_direction_followthrough_present` simultaneously drops from `13 / 24` to `10 / 24` on the same mainline comparison surface. The reviewed `NRR vs Ablation` read remains auxiliary boundary evidence only: Gemini reaches support there, Anthropic remains inconclusive, and the Anthropic-side auxiliary comparison keeps `Baseline` stronger than `NRR` on `appropriate`. This surface is therefore carried forward only as a bounded downstream-boundary result, not as a pair-wide downstream guarantee, explicit quarantine closure, or later-horizon stability claim.

The auxiliary repaired read uses a narrower direct trace than the mainline protocol-era fields: `accepted_branch_followthrough_present` maps to `accepted_direction_followthrough_present`, `reopened_choice_absent` plus `alternative_branch_push_absent` jointly trace the manuscript's rejected-direction non-intrusion requirement, and that mapping is kept explicit in the bundled closure import note rather than treated as a one-to-one field replacement.

5.3 Side-Topic / Quarantine Supporting Surface

Because this surface depends on a local repaired `baseline vs revised Energy policy` comparison rather than on the paper-level mainline `baseline vs Energy policy` contract, it is carried here as a bounded supporting surface and not as a second manuscript supported claim unit.

The repaired first-step side-topic / quarantine supporting surface is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)

- family scope: the repaired first-step side-topic / quarantine family only
- horizon scope: the first accepted side-topic workstart checkpoint only
- observability scope: `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`
- comparison scope: matched baseline vs revised Energy-policy under the repaired first-step side-topic / quarantine pilot conditions
- supported behavior: clean first-step side-topic workstart under explicit side-first acceptance
- explicit failure boundary: the gain is incremental rather than a rescue from total baseline failure, and the main lift over baseline is concentrated in `C_incident_response`; because this read uses a local repaired comparison rather than the inherited paper-level contract, it is carried as supporting evidence rather than as a manuscript supported claim unit
- unsupported extension: no provider-universal side-topic commitment, uniform template lift, later- horizon side-topic stability claim, or upgrade to a second manuscript supported claim is authorized

In manuscript form, the bounded supporting read is:

> Under the tested provider pair, for the repaired first-step side-topic / quarantine family, at the first accepted side-topic workstart checkpoint, and on the directly scored fields `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`, the revised Energy-policy arm is supported relative to matched baseline only for clean first-step side-topic workstart under explicit side-first acceptance. This surface is carried only as bounded supporting evidence because its comparison contract is the repaired local `baseline vs revised Energy policy` surface rather than the manuscript's mainline `baseline vs Energy policy` contract. It excludes provider-universal side-topic commitment, uniform template lift, and later-horizon side-topic stability, and it remains bounded by the fact that the main incremental lift over baseline is concentrated in `C_incident_response`.

6 Evidence Sources

The scientific argument in this paper draws on three evidence layers. Supporting texts provide the inherited program design, policy, dynamics,

and protocol vocabulary used here. Analysis notes document the readings used for the early ambiguity-turn claim, the downstream-acceptance boundary, the repaired first-step side-topic / quarantine result, the later-horizon boundary, and the policy/dynamics interpretation layer. An imported reviewed readout preserves the repaired `NRR vs Ablation` downstream comparison as auxiliary boundary evidence only.

Detailed filenames are listed in `reproducibility.md` and in the supporting evidence map rather than reproduced in the main text. That separation keeps the paper focused on the scientific interpretation while still making the evidence traceable.

The supporting bundle records provenance and inherited dependencies, but it is not the sole basis for every result reported here. In particular, the older downstream-acceptance memo preserved in `supporting_layer_sources/current/` remains a pre-repair historical read. The downstream-acceptance section instead relies on the closure import note, the boundary note, the baseline-vs-Energy comparison memo with its four run-annotation CSVs, and the imported reviewed auxiliary comparison listed above. The same evidence bundle also includes the repaired side-topic comparison memo with its four run-annotation CSVs, together with the later-horizon two-provider memo and scored read notes needed to inspect that boundary in detail.

6.1 First Evidence Import

The strongest current entry point for the `Guarantee` line is the frozen early ambiguity-turn pilot. Under matched baseline vs Energy-policy conditions, both tested provider slices moved from `0 / 24` to `24 / 24` on `resolution_attempt_present`, from `8 / 24` to `24 / 24` on `unilateral_commit_absent`, and from `0 / 24` to `24 / 24` on `appropriate`. Baseline failure modes remained visible as either ambiguity-preserving but unresolved responses or unilateral/content-before-resolution behavior. The carry-forward read is therefore that the Energy-policy surface supports non-unilateral priority resolution at the early decision boundary on the tested provider pair. This is a bounded early-control claim only; it does not by itself establish downstream followthrough, quarantine behavior, later-horizon stability, or provider-universal validity.

6.2 Second Evidence Import

The downstream-acceptance boundary is the current second `Guarantee` entry point. The inherited paper-level comparison contract stays on matched `baseline vs Energy policy` conditions, where both tested provider slices move from `0 / 24` to `24 / 24` on `preacceptance_rework_burden_absent`. That shared gain is the current supported downstream-boundary behavior. The residual remains visible in the same mainline surface: Gemini still shows partial first accepted-direction followthrough on part of the tested rows, so the claim does not close as pair-wide downstream-quality support. The imported reviewed readout then keeps a narrower auxiliary repaired comparison visible directly inside this repository: on reviewed `NRR vs Ablation`, Gemini reaches support, Anthropic remains inconclusive, and the same reviewed memo keeps Anthropic `Baseline` stronger than Anthropic `NRR` on `appropriate`. The truthful carry-forward read is therefore bounded support for preacceptance burden removal under the mainline comparison, with the repaired `NRR vs Ablation` surface used only as auxiliary boundary evidence. This surface does not yet license pair-wide downstream guarantee language, explicit quarantine closure, later-horizon stability claims, or any claim that the auxiliary repaired read replaces the paper-level comparison contract.

6.3 Third Evidence Import

The third `Guarantee` entry point is the repaired first-step side-topic / quarantine supporting result. Under matched baseline vs revised Energy-policy conditions, the revised arm reached `appropriate = 12 / 12` in both provider slices, but the honest read is incremental rather than global: baseline was already fairly strong on `A_trip_planning` and `B_hiring_packet`, and the main lift over baseline concentrated in `C_incident_response` (`1 / 4 -> 4 / 4` on Anthropic and `0 / 4 -> 4 / 4` on Gemini). Because this surface sits on a local repaired `baseline vs revised Energy policy` comparison rather than on the paper-level mainline contract, the carry-forward read is a bounded first-step side-topic workstart supporting surface under explicit side-first acceptance, not a second manuscript supported claim and not a universal side-topic/quarantine or later-horizon stability claim.

Behavior-First Manipulation Check A separate provider-pair rerun on the repaired behavior-first surface does not enter this paper as a

supported `Guarantee` claim unit or as a boundary surface. Its narrower manuscript role is to show that, on the tested provider pair, repaired opening-comparison insertion produces a readable change in first-move behavior: implementation responses mostly remained in direct relation work, whereas ablation responses mostly opened with comparison-heavy framing. This result is therefore carried here only as bounded behavioral-separation evidence. It does not establish a clean two-stage realization: the ablation arm still most often opens with comparison framing and then continues with a follow-up question rather than separating those moves cleanly, Anthropic `D2` remains a `1 / 10` collapse row, and the rerun does not establish pair-level continuation-quality superiority. Its role in this manuscript is therefore a secondary bounded note inside the evidence-import discussion rather than a main claim unit or boundary surface.

Formal Carried Surfaces The formal scientific surfaces carried by this manuscript are therefore three comparison-backed carried surfaces plus one diagnostic later-horizon boundary:

- Early ambiguity-turn supported claim: tested provider pair, early decision boundary, and direct observability on `resolution_attempt_present`, `unilateral_commit_absent`, and `appropriate`.
- Downstream-acceptance boundary: tested provider pair, first accepted- direction downstream boundary, and direct observability on `accepted_direction_followthrough_present`, `rejected_direction_intrusion_absent`, `preacceptance_rework_burden_absent`, and `appropriate`.
- First-step side-topic / quarantine supporting surface: tested provider pair, first accepted side-topic workstart checkpoint, and direct observability on `chosen_side_direction_workstart_present`, `suspended_main_reference_absent`, `main_direction_intrusion_absent`, and `appropriate`; this surface remains supporting evidence because it depends on a local repaired `baseline vs revised Energy policy` comparison.
- Later-horizon failure boundary: tested provider pair, measured later checkpoints especially `turn9`, and relation-preservation failure surfaces under a diagnostic one-arm later-checkpoint read.

The behavior-first rerun remains outside this formal surface list and is carried only as a secondary bounded behavioral-separation note.

6.4 Policy/Dynamics Interpretation Layer

Taken together, these carried surfaces define a bounded progression rather than a universal recipe. They share one disclosure schema, but not one uniform evidentiary contract: the first three surfaces are comparison-backed, while the later-horizon boundary is a diagnostic later-checkpoint read. At the earliest boundary, the supported behavior is `defer_with_resolution` and the corresponding dynamics motif is *Stable Resolution*. At the downstream acceptance boundary, the supported behavior is bounded burden removal under matched `baseline vs Energy policy` conditions, but Gemini still retains first-followthrough residuals and the auxiliary repaired `NRR vs Ablation` read remains mixed, so this surface cannot yet be read as pair-wide *Useful Continuation Hold*. At the repaired first-step side-topic supporting surface, the carried behavior is a clean transition from `accepted_direction` into `active_objective` with `suspended_objective` still held in quarantine, which corresponds to *Clean Quarantine Hold*. Because that surface depends on a local repaired comparison, it remains supporting evidence rather than the paper's second supported claim. The later-horizon limit then becomes easier to state precisely: the first broad break is not simple suspended-main reactivation but weakening *Relation-Preservation Hold* until *Premature Relation Collapse* appears. Table 2 summarizes the provider-specific residual structure that keeps the current `Guarantee` claims bounded.

7 Later-Horizon Failure Boundary

The later-horizon boundary surface is:

- provider scope: the tested provider pair only (`claude-sonnet-4-6` and `gemini-2.0-flash`)
- family scope: the frozen `priority-resolution horizon-stability` family only, separate from the repaired first-step side-topic/quarantine family and used to read the later continuation of the carried side-topic line
- horizon scope: later checkpoints `turn7`, `turn9`, and `turn11` only, with `turn9` the first shared clean failure checkpoint

- observability scope: `chosen_side_direction_progress_present`, `user_detail_use_present`, `unresolved_relation_preserved_or_narrowly_checked`, `suspended_main_reference_absent`, `starter_question_repeat_absent`, and `appropriate`
- comparison scope: not a matched comparison fill; diagnostic one-arm later-checkpoint read over the provider-scored annotation tables for the active later-horizon family
- supported behavior: boundary-only detection that relation preservation weakens before suspended- main quarantine breaks, with the limited safe hold that the first side-topic switch can survive into the next checkpoint
- explicit failure boundary: later degradation remains real, template-dependent, and provider-sensitive; the clearest shared failure surface is `C_incident_response` at `turn9`, where both providers collapse the unresolved relation by prematurely committing an ordering between `impact` and `root cause`
- unsupported extension: no matched-comparison claim, no fourth supported `Guarantee` claim, no provider-universal or family-universal degradation depth, no long-horizon stability guarantee, and no claim that prompt-layer control alone secures stable `turn11` behavior

In manuscript form, the bounded later-horizon boundary read is:

> Under the tested provider pair, for the frozen `priority-resolution horizon-stability` family, kept separate from the repaired first-step side-topic/quarantine family but used to read the later continuation of the carried side-topic line, at later checkpoints `turn7`, `turn9`, and `turn11`, and on the directly scored later-horizon observability fields listed above, the current later-horizon surface is carried only as a diagnostic one-arm boundary read, not as a matched-comparison fill. The limited safe hold is that the first side-topic switch can survive into the next checkpoint while suspended-main quarantine remains visible, but later degradation remains real, template-dependent, and provider-sensitive. The clearest shared failure surface is the `C_incident_response` template at `turn9`, where both providers collapse the still unresolved relation by prematurely committing an ordering between `impact` and `root cause`. The correct read is therefore that later-horizon instability emerges through weakening relation preservation before suspended-main quarantine itself breaks. This later-horizon material is a boundary on the current supported surfaces only: it does not authorize a matched-comparison claim, a fourth supported `Guarantee` claim, provider-universal degradation depth, or long-horizon stability language.

8 Reproducibility

The materials retained for this paper are intentionally narrow:

- a current TeX source
- a current PDF snapshot
- the current figure assets used by that TeX source
- a supporting-layer evidence map
- bounded claim, supporting, and boundary drafting memos
- an imported reviewed downstream-acceptance readout for the current downstream boundary
- a secondary behavior-first manipulation-check note
- a later-horizon failure-boundary memo
- a policy/dynamics manuscript-integration memo
- bundled supporting-layer source files
- stable build and verification wrappers

These materials expose both the paper and the specific sources used to justify its bounded conclusions. Checksum verification runs on the tracked artifact set, and the documented build wrapper can produce a PDF in this environment through a source-rendered fallback when cached TeX resources are incomplete. Full typeset reproduction therefore remains environment-dependent here, but the tracked materials are organized so that the bounded claim, the reviewed downstream-boundary readout, and the supporting sources can be inspected without relying on workspace-only links.

9 Discussion

The non-evaluative principle should enter this line as bounded engineering discipline rather than theorem inflation. In the current read, the practical issue is not simply whether unresolved content exists, but whether later interaction forces premature relation collapse while the interaction is still in a retained state. The `Guarantee` paper is the correct place to express that read as a bounded downstream claim surface. In that sense, the manuscript

is adjacent to prior mixed-initiative, clarification, and grounding work, but its objective is different: it does not primarily propose a better question-asking heuristic or a general dialogue-success policy. Instead, it asks which bounded commit/defer claims remain supportable under explicit provider, family, horizon, and observability restrictions [8, 9, 11, 14, 15, 16, 17].

That downstream read should not be reduced to a simple rule of "never recommend one next move." A response may advance with a provisional recommendation and still preserve the user's live context if the governing constraint structure remains visible or explicitly marked as contingent. The harder failure is premature interpretation fixation: proceeding as if one local reading of the user's still-live objective, satisfaction condition, or constraint ordering had already been settled. Framed that way, unresolved-relation preservation is not mere hesitation. It is bounded context retention under a visible interaction state.

This paper also remains deliberately narrow. It is based on two tested provider slices, one supported early-control claim, one explicit downstream boundary, one side-topic supporting surface, and one later-horizon boundary read rather than on a broad cross-provider or long-horizon guarantee. Those limits are not defects to be hidden: they are part of the bounded-claim discipline that keeps the paper from overstating what the current evidence can support.

This staging matters for how the results should be read. The paper is not claiming a monotone march from early control support to pair-wide downstream success. The truthful sequence is narrower: the early ambiguity-turn surface is clean on the tested provider pair, while the repaired downstream-acceptance boundary supports only bounded preacceptance-burden removal under matched `baseline vs Energy policy` conditions, while Gemini still shows `accepted_direction_followthrough_present` dropping from `13 / 24` to `10 / 24` and the auxiliary repaired `NRR vs Ablation` read remains mixed, with Anthropic still `inconclusive`, Gemini reaching `support`, and Anthropic `Baseline` remaining stronger than Anthropic `NRR` on `appropriate`. On that auxiliary repaired surface, the Anthropic gap is large rather than cosmetic: `Baseline` reaches `142 / 144` on `appropriate`, whereas `NRR` reaches `55 / 144`. That auxiliary asymmetry should remain visible because it marks a boundary on the repaired comparison surface, not because it upgrades the paper-level read. The paper-level downstream interpretation remains carried by the matched `baseline vs Energy policy` surface only: bounded support for preacceptance-burden removal is visible there, while pair-wide downstream- quality closure is not. Accordingly, the auxiliary repaired read is kept as boundary evidence of

unresolved downstream fragility rather than as the authorizing basis for a paper-level early-control / downstream-quality trade-off claim.

The repaired behavior-first rerun sharpens that boundedness rather than relaxing it. Its correct manuscript role is a secondary manipulation-check note inside the evidence-import discussion: it shows that opening-comparison insertion changes how the next move is framed on the tested provider pair. But that empirical separation should not be overread as a broader downstream- quality result. The manipulated form remains compressed, one clear collapse row remains visible, and the current evidence still falls short of pair-level continuation-quality superiority. Keeping that limitation adjacent to the reported rerun result is part of the same bounded-claim discipline rather than an afterthought.

These results should also be read as a reusable bounded test rather than as the last word on guarantee behavior. What is fixed here is not an exhaustive operator set or a closed theorem, but a bounded way to keep later work honest: explicit supported claims, explicit supporting surfaces, explicit boundary surfaces, explicit comparison surfaces, provider-sliced reporting, and visible failure boundaries. That structure is meant to make follow-on work easier. Later readers should be able to rerun these guarantee questions under new providers, new downstream families, new operator variants, and longer horizons without having to guess which comparison contract or boundary rule the current manuscript relied on.

10 Limitations

The paper is limited to a narrow tested-provider surface: two provider slices, one supported early-control claim, one explicit downstream boundary, one side-topic supporting surface, and one later-horizon boundary read rather than a broad cross-provider or long-horizon guarantee.

The downstream evidence also remains non-uniform. The repaired downstream- acceptance boundary supports bounded preacceptance-burden removal under matched `baseline vs Energy policy` conditions, but Gemini still shows reduced `accepted_direction_followthrough_present`, and the auxiliary repaired `NRR vs Ablation` surface remains mixed, with Anthropic still `inconclusive`. The behavior-first rerun likewise remains secondary evidence only and does not establish pair-level continuation-quality superiority. These results should therefore be read as a bounded test rather than as an exhaustive guarantee theorem or final cross-provider closure.

11 Conclusion

This paper establishes a bounded `Guarantee` claim. It reports one supported early ambiguity-turn result, carries the downstream-acceptance comparison as an explicit boundary with bounded support for `preacceptance_rework_burden_absent` under matched `baseline vs Energy policy` conditions, keeps the repaired `NRR vs Ablation` comparison as an auxiliary boundary comparator, reports the repaired first-step side-topic/quarantine result only as supporting evidence under matched `baseline vs revised Energy policy` conditions, and keeps the later-horizon limit explicit as boundary reporting rather than as an additional supported claim. The result is closer to a bounded verification or assurance statement over explicit observables than to a universal correctness claim [18, 19]. Early control support does not automatically extend into pair-wide downstream acceptance, the repaired downstream boundary remains limited to burden removal rather than pair-wide downstream quality, and the Anthropic-side auxiliary comparison keeps `Baseline` stronger than `NRR` on `appropriate`. Broader provider coverage, broader downstream family coverage, and later-horizon closure remain outside the present claim.

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Surface	**Anthropic-side read**	**Gemini-side read**	**Boundary implication**
Early ambiguity-turn	Clean support on the tested early surface	Clean support on the tested early surface	Safe to state only a bounded early-control claim, not broader downstream closure
Downstream acceptance boundary	Mainline burden-removal support is clean; auxiliary repaired read stays `inconclusive` and keeps `Baseline` stronger than `NRR` on `appropriate`	Mainline burden-removal support is present, but `accepted_direction_followthrough_present` drops from `13 / 24` to `10 / 24`; auxiliary repaired read reaches `support`	Keep the claim at bounded burden-removal support under the mainline contract; use the repaired read only as boundary evidence, not as a replacement comparison rule
First-step side-topic / quarantine	Lift is visible, with strongest incremental gain on `C_incident_response`	Lift is also visible, again with the main gain concentrated on `C_incident_response`	Treat the result as template-bounded improvement, not uniform side-topic stability
Later-horizon boundary	Degradation remains visible at later checkpoints	Degradation remains visible at later checkpoints	Preserve later-horizon text as boundary reporting rather than a fourth supported guarantee

Table 2: Provider-specific residual structure that keeps the current `Guarantee` claims bounded.